How I would break this

1. Crash points table

All locations below are under src/. The frame is the commit unit; the optional final 00 byte is expendable padding. Named hooks kill the process with SIGKILL. See docs/GAP_FIX_REPORT.md for seeds and counts.

Hook and exact function	Disk at kill	Next open
init_before_rename: engine._create	Synced .init, no authoritative path yet	Remove temp; create header again
before_write: engine._write	Prior frames and guard	New operation absent
torn_write: engine._write	Incomplete frame overwrote old guard	Drop incomplete tail; restore guard
after_write: engine._write	Complete frame plus guard; not synced	Replay full CRC-valid frame
after_fsync: engine._write	Frame and guard synced; RAM stale	Rebuild index; include operation
after_return: serve.run/execute	Engine returned; ACK not printed	Include durable operation
recovery_mid_scan: engine._scan	File unchanged; RAM incomplete	Restart replay
recovery_before_truncate: engine._recover	Torn tail remains	Repeat repair
recovery_after_truncate: engine._recover	Valid prefix and guard synced	Replay clean file
compact_mid_copy: engine._copy_live	Old authority plus partial temp	Use old file; remove temp
compact_before_rename: engine._compact_locked	Old file plus synced new temp	Use old file; remove temp
compact_after_rename: engine._compact_locked	Path names complete new file	Recover new file
compact_after_swap: engine._compact_locked	New file/directory synced; handle swapped	Recover new file; append to it
close_mid: engine.close	Accepted frames already durable	Recover without close cleanup

2. Ordering argument

Under the process-death model, os.pwrite loops until the frame is complete, then writes one EOF guard; os.fsync returns before index publication, engine success and serve ACK. The next append overwrites the guard, not earlier frame bytes. A partial frame is discarded as a whole. Removing one trailing byte from an acknowledged file removes only the guard. The guard is not a commit marker.

Recovery uses os.ftruncate, restores the guard when space permits, then os.fsync. Compaction copies live data and guard, fsyncs/closes the temp, calls os.replace, fsyncs the parent directory, then reopens/swaps the descriptor and index. Directory fsync also follows initial-file replacement. Atomic rename is assumed on the tested Linux ext4 filesystem.

On append failure, rollback truncates the attempted append, restores the old guard and syncs. The handle is poisoned. Failed rollback reports UNCERTAIN: reopen and reconcile before retrying; absence is not promised.

3. Known weaknesses

1. Replay is O(log size). Million-record measurements pass on this laptop, but a sufficiently long log, slower storage or different CPU load can exceed two seconds.

2. One mutex blocks readers/writers during compaction. Encoding is chunked, but retained input values, queued batches and the index still consume RAM. The 10,000 distinct 1 MiB value combination is not qualified.

3. Evidence covers process kills on ext4, not power loss, lying hardware, NTFS or arbitrary middle-file damage. CRC32 is not cryptographic authentication.

4. Legacy files at a cap with no spare byte remain readable without migration growth. Their unguarded tails need room and a current write/compaction before the guard guarantee applies. Older engines can remove the guard.

5. OS scheduling is not reproduced exactly by a seed. Full transcripts and deterministic hooks supplement seeded plans. The exact shared-batch oracle has a 250,000-state budget; exhaustion is a tool error, never PASS.

4. What we tried that failed

A zero-progress pwrite once hung the engine; a progress guard now rejects it. Early-ACK and partial-batch planted defects are detected by negative regressions. Skipped tail truncation previously escaped random kills; deliberate torn-write trials now expose it. Invalid UTF-8 once killed serve; malformed lines now get errors. The literal one-byte test previously lost acknowledged data; the EOF guard fixes that without accepting partial frames. A test also incorrectly required global reply order across writers; it now checks exact responses without that unsupported order. Raw pre-fix evidence is retained.

5. If we had 24 more hours

1. Add a byte-based admission budget for retained batch inputs/queues; stress distinct maximum-sized values under a stated RAM budget.

2. Add a crash-safe checkpoint or bounded replay strategy, then benchmark 10 million records and cache/load variations with a hard two-second gate.

3. Compare the concurrent oracle against a separate bounded reference model, and expand adversarial shared-key histories without silently raising its search budget.

The original prompt log remains pending by the user's choice. Physical Wi-Fi-off rehearsal and live source-line answers need human sign-off; automated counts do not certify them. Printable two-page layout: output/pdf/BREAKING.pdf.